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Chapter 1

Background: Finite Groups, Representations, Intertwiners

1.1 Hypotheses and Notation

Definition 1 (Standing hypotheses). Throughout the blueprint the symbol G denotes a finite group: in Lean,

$$[\text{Group } G] \text{ } [\text{Fintype } G].$$

The symbol k denotes an algebraically closed field whose characteristic does not divide $|G|$:

$$[\text{Field } k] \text{ } [\text{IsAlgClosed } k] \text{ } [\text{Invertible } ((\text{Fintype.card } G : k))].$$

We write $|G|$ for $\text{Fintype.card } G$ and view it as an element of k via the canonical map $\mathbb{Z} \rightarrow k$.

1.2 Representations and Their Bundlings

Definition 2 (k -linear representation). A k -linear representation of G on a k -module V is a monoid homomorphism $\rho : G \rightarrow (V \rightarrow_{\ell[k]} V)^\times$ from G into the multiplicative monoid of k -linear endomorphisms of V . In Mathlib this is the abbreviation $\text{Representation } k \ G \ V := G \rightarrow^* (V \rightarrow [k] \ V)$.

Definition 3 ($\rho.\text{asModule} : \text{MonoidAlgebra-module structure}$). Given $\rho : \text{Representation } k \ G \ V$, the type .asModule carries a canonical $\text{MonoidAlgebra } k \ G$ -module structure that extends the k -action on V . Conversely, any $k[G]$ -module gives rise to a representation. This identification (Mathlib records it as the equivalence $\text{Rep.equivalenceModuleMonoidAlgebra}$) is the bridge between representation-theoretic and module-theoretic statements.

Definition 4 (Bundled categories $\text{Rep } k \ G$ and $\text{FDRep } k \ G$). $\text{Rep } k \ G$ is the category of k -linear G -representations: objects are pairs (V, ρ) as above, morphisms are G -equivariant k -linear maps. $\text{FDRep } k \ G$ is the full subcategory of finite-dimensional representations. For $V : \text{FDRep } k \ G$, the underlying vector space is $V.V$ (or by coercion just V) and the action is $V. : G \rightarrow^* (V \rightarrow [k] \ V)$.

Definition 5 (Intertwining map = morphism in $\text{FDRep } k \ G$). An *intertwining map* (or G -map) from V to W in $\text{FDRep } k \ G$ is a morphism $V \rightarrow W$, i.e. a k -linear map $T : V \rightarrow W$ such that $T \circ \rho_V(g) = \rho_W(g) \circ T$ for all $g \in G$. The hom space $V \rightarrow W$ is finite-dimensional over k .

Definition 6 (Isomorphism of representations). Two representations $V, W : \mathbf{FDRep} \ k \ G$ are *isomorphic*, $V \cong W$, if there is an isomorphism $V \cong W$ in the category $\mathbf{FDRep} \ k \ G$: equivalently, a k -linear isomorphism whose underlying map is G -equivariant.

Definition 7 (Subrepresentation, invariants). A *subrepresentation* of ρ on V is an $(\mathbf{MonoidAlgebra} \ k \ G)$ -submodule of `.asModule`; equivalently, a k -subspace $W \subseteq V$ with $\rho(g)(W) \subseteq W$ for every $g \in G$. The G -invariants are $V^G := \{v \in V : \rho(g)v = v \ \forall g\}$, a k -subspace `Representation.invariants`.

Definition 8 (Direct sum of representations). The category $\mathbf{FDRep} \ k \ G$ has biproducts: for $V, W : \mathbf{FDRep} \ k \ G$ there is $V \oplus W : \mathbf{FDRep} \ k \ G$ (notation $V \oplus W$) on the underlying vector space $V \oplus W$ with the diagonal action.

Definition 9 (Irreducible representation). A finite-dimensional representation $V : \mathbf{FDRep} \ k \ G$ is *irreducible* (or *simple*) if `CategoryTheory.Simple V`: V is non-zero and every subrepresentation is either 0 or V itself. In Lean this is provided as a typeclass `[CategoryTheory.Simple V]`.

Definition 10 (Inner product on k^G). For $f, h : G \rightarrow k$ define

$$\langle f, h \rangle_G := \frac{1}{|G|} \sum_{g \in G} f(g) h(g^{-1}).$$

When $h = \overline{h'}$ is replaced by complex conjugation in the unitary case, this recovers the usual Hermitian inner product. We use the algebraic form throughout, which agrees with Mathlib's character formula `Representation.card_inv_mul_sum_char_mul_char_eq_finrank`.

Chapter 2

Maschke's Theorem and Semisimplicity

The strategy is the standard “averaging trick”: start from any k -linear retraction of a G -equivariant inclusion, then average over G to get a G -equivariant retraction. We follow the Mathlib formulation in `Mathlib.RepresentationTheory.Maschke`.

2.1 The Conjugate of a Linear Map

Definition 11 (Conjugate of a linear map). Let V, W be $k[G]$ -modules and $\pi : W \rightarrow V$ a k -linear map. For each $g \in G$ define the *conjugate*

$$(\pi^g)(v) := g^{-1} \cdot \pi(g \cdot v),$$

i.e. `.conjugate g v = (single g ⁻¹ 1) • ((single g 1) • v)`.

Lemma 12 (Conjugate fixes equivariant inclusions). *If $i : V \rightarrow W$ is $k[G]$ -linear and $\pi \circ i = \text{id}_V$, then for every $g \in G$ also $\pi^g \circ i = \text{id}_V$ (as k -linear maps).*

Proof. By unfolding: $\pi^g(i(v)) = g^{-1} \cdot \pi(g \cdot i(v)) = g^{-1} \cdot \pi(i(g \cdot v)) = g^{-1} \cdot (g \cdot v) = v$, where we used $k[G]$ -linearity of i and the retraction hypothesis. This is `LinearMap.conjugate_i` in Mathlib. $\square$

2.2 The Averaged (Equivariant) Projection

Definition 13 (Sum-of-conjugates). Define $\Sigma\pi := \sum_{g \in G} \pi^g$. The Mathlib name is `LinearMap.sumOfConjugatesEquivariant`, and the corresponding pointwise formula is `sumOfConjugatesEquivariant_apply`.

Definition 14 (Equivariant projection). Set $\bar{\pi} := |G|^{-1} \Sigma\pi$. The Mathlib name is `LinearMap.equivariantProjection` and pointwise `equivariantProjection_apply`: $\bar{\pi}(v) = \frac{1}{|G|} \sum_g \pi^g(v)$.

Lemma 15 ($\bar{\pi}$ is $k[G]$ -linear). *The map $\bar{\pi} : W \rightarrow V$ is $k[G]$ -linear: it commutes with the action of every group element.*

Proof. Bundled into the Mathlib definition: `LinearMap.equivariantProjection` has codomain $(W \rightarrow [\text{MonoidAlgebra } k \ G] \ V)$. The proof in Mathlib is by reindexing the sum $\sum_g \pi^g$: conjugating by h permutes the summands ($g \mapsto h^{-1}g$) and the prefactor $|G|^{-1}$ is fixed. $\square$

Lemma 16 ($\bar{\pi}$ retracts i). *If $i : V \rightarrow W$ is $k[G]$ -linear and $\pi \circ i = \text{id}_V$, then $\bar{\pi} \circ i = \text{id}_V$.*

Proof. Compute, for $v \in V$: $\bar{\pi}(i(v)) = |G|^{-1} \sum_g \pi^g(i(v)) = |G|^{-1} \sum_g v = |G|^{-1} \cdot |G| \cdot v = v$, using Lemma 12 for the second equality and the invertibility of $|G|$ for the last. This is `LinearMap.equivariantProjection_condition`. $\square$

2.3 Maschke's Theorem

Theorem 17 (Maschke: every $k[G]$ -injection splits). *Let V, W be $k[G]$ -modules and $f : V \rightarrow W$ a $k[G]$ -linear injection. Then f admits a $k[G]$ -linear left inverse: there exists $g : W \rightarrow V$ in $k[G]\text{-Mod}$ with $g \circ f = \text{id}_V$.*

Proof. Choose any k -linear retraction $\pi : W \rightarrow V$ of f (it exists because $V \rightarrow W$ is an injection of k -vector spaces). Take $g := \bar{\pi}$ from Definition 14; it is $k[G]$ -linear (Lemma 15) and retracts f (Lemma 16). This is the proof of `MonoidAlgebra.exists_leftInverse_of_injective`. $\square$

Theorem 18 (Maschke: every submodule has a complement). *For any $k[G]$ -module V and any $k[G]$ -submodule $p \subseteq V$ there is a $k[G]$ -submodule $q \subseteq V$ with $V = p \oplus q$ (i.e. `IsCompl p q`).*

Proof. Apply Theorem 17 to the inclusion $p \hookrightarrow V$ to obtain a $k[G]$ -linear retraction $g : V \rightarrow p$. Then $q := \ker g$ is a $k[G]$ -submodule with $p \oplus q = V$. $\square$

Theorem 19 (Maschke: V is semisimple as a $k[G]$ -module). *For every $k[G]$ -module V we have `IsSemisimpleModule (MonoidAlgebra k G) V`: every submodule has a complement, equivalently, V is a direct sum of simple submodules.*

Proof. This is the Mathlib instance `MonoidAlgebra.Submodule.instIsSemisimpleModule`, which packages Theorem 18 into the typeclass `IsSemisimpleModule`. $\square$

Corollary 20 (Every f.d. rep is a direct sum of irreducibles). *Every $V : \text{FDRep } k \ G$ is isomorphic to a finite direct sum of simple objects of $\text{FDRep } k \ G$.*

Proof. By Theorem 19, $V.asModule$ is semisimple. A finitely generated semisimple module is a finite direct sum of simple modules; transporting along the equivalence `Rep.equivalenceModuleMonoidAlgebra` gives the statement in `FDRep k G`. $\square$

Chapter 3

Schur's Lemma and Matrix-Coefficient Orthogonality

3.1 Schur's Lemma

Theorem 21 (Schur's Lemma — dimension form). *Let $V, W : \mathbf{FDRep} \ k \ G$ both be simple. Then*

$$\dim_k(V \rightarrow W)_G = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. This is `FDRep.finrank_hom_simple_simple`. The proof packaged in Mathlib uses the more general `CategoryTheory.Schur` for an abelian k -linear category with simple objects and k algebraically closed: any nonzero morphism $T : V \rightarrow W$ between simples has zero kernel and zero cokernel by simplicity of V, W , hence is an iso; for endomorphisms $T : V \rightarrow V$, since k is algebraically closed T has an eigenvalue λ , and $T - \lambda \text{id}$ is a non-iso endomorphism of a simple object, hence zero. $\square$

Corollary 22 (Schur scalar form). *For $V : \mathbf{FDRep} \ k \ G$ simple, every G -equivariant endomorphism of V is a scalar multiple of id_V . The ring $\text{End}_G(V)$ is isomorphic to k .*

Proof. By Theorem 21 with $V = W$, $\dim_k \text{End}_G V = 1$. Since $\text{id}_V \in \text{End}_G V$ is nonzero, it spans, so every endomorphism is a scalar. $\square$

3.2 The Symmetrisation $A \mapsto |G|^{-1} \sum_g \rho(g) A \rho(g)^{-1}$

The following few lemmas isolate the averaging argument used in the matrix-coefficient orthogonality calculation. They are entirely elementary and do not yet appear as standalone Mathlib declarations.

Definition 23 (G -symmetrisation of a linear map). Given $\rho : \text{Representation } k \ G \ V$, $\sigma : \text{Representation } k \ G \ W$ and a k -linear $A : V \rightarrow W$, define

$$\text{Sym}_{\rho, \sigma}(A) := \frac{1}{|G|} \sum_{g \in G} \sigma(g) A \rho(g)^{-1} \in (V \rightarrow_{\ell[k]} W).$$

Lemma 24 (Symmetrisation lands in equivariant maps). *$\text{Sym}_{\rho,\sigma}(A)$ is G -equivariant: it intertwines ρ and σ .*

Proof. Compute, for $h \in G$:

$$\begin{aligned}\sigma(h) \text{Sym}(A) &= \frac{1}{|G|} \sum_g \sigma(h) \sigma(g) A \rho(g)^{-1} = \frac{1}{|G|} \sum_g \sigma(hg) A \rho(g)^{-1} \\ &= \frac{1}{|G|} \sum_{g'} \sigma(g') A \rho(h^{-1}g')^{-1} \quad (g' := hg) \\ &= \frac{1}{|G|} \sum_{g'} \sigma(g') A \rho(g')^{-1} \rho(h) = \text{Sym}(A) \rho(h).\end{aligned}$$

□

Lemma 25 (Symmetrisation between non-isomorphic simples is zero). *If $V, W : \text{FDRep } k \text{ } G$ are simple and not isomorphic, then $\text{Sym}_{V.\rho, W.\rho}(A) = 0$ for every k -linear $A : V \rightarrow W$.*

Proof. By Lemma 24, $\text{Sym}(A)$ is an intertwiner $V \rightarrow W$. By Theorem 21, the space of such intertwiners has dimension 0 since $V \not\cong W$, hence $\text{Sym}(A) = 0$. □

Lemma 26 (Symmetrisation on a single simple is a scalar). *Let $V : \text{FDRep } k \text{ } G$ be simple. For every $A : V \rightarrow V$ (k -linear), there is a unique scalar $c(A) \in k$ such that $\text{Sym}_{V.\rho, V.\rho}(A) = c(A) \text{id}_V$. Furthermore*

$$c(A) = \frac{\text{tr}(A)}{\dim_k V}.$$

Proof. By Lemma 24 the symmetrisation is in $\text{End}_G V$, which by Corollary 22 is one-dimensional spanned by id_V . Hence $\text{Sym}(A) = c(A) \text{id}_V$ for some $c(A) \in k$. Take the k -trace of both sides: the left side has trace $\frac{1}{|G|} \sum_g \text{tr}(\rho(g) A \rho(g)^{-1}) = \text{tr}(A)$ (cyclicity of trace), and the right side has trace $c(A) \dim_k V$. So $c(A) = \text{tr}(A) / \dim_k V$. □

3.3 Matrix-Coefficient Orthogonality (Schur II)

Theorem 27 (Matrix-coefficient orthogonality, single irrep). *Let $V : \text{FDRep } k \text{ } G$ be simple of dimension $d := \dim_k V$, and let $\{e_1, \dots, e_d\}$ be a k -basis of V . Write $\rho_{ij}(g)$ for the (i, j) -th matrix coefficient of $V.\rho(g)$ in this basis. Then*

$$\frac{1}{|G|} \sum_{g \in G} \rho_{ij}(g) \rho_{kl}(g^{-1}) = \frac{\delta_{il} \delta_{jk}}{d}.$$

Proof. Apply Lemma 26 to the rank-one k -linear map $A := E_{lj}$ (the matrix unit sending e_j to e_l , all other basis vectors to 0). Then $\text{tr}(A) = \delta_{lj}$, so $c(A) = \delta_{lj}/d$, and reading off the (i, k) -th matrix entry of $\text{Sym}(A) = c(A) \text{id}_V$ on both sides gives $\frac{1}{|G|} \sum_g \rho_{ij}(g) \rho_{kl}(g^{-1}) = \delta_{il} \delta_{jk} / d$. □

Theorem 28 (Matrix-coefficient orthogonality, distinct irreps). *Let $V, W : \text{FDRep } k \text{ } G$ be simple with $V \not\cong W$, and choose bases of V and W . Then for all matrix indices,*

$$\frac{1}{|G|} \sum_{g \in G} \rho_{ij}(g) \sigma_{kl}(g^{-1}) = 0,$$

where $\rho = V.\rho$ and $\sigma = W.\rho$.

Proof. Apply Lemma 25 to any rank-one $A : V \rightarrow W$ and read off the result in coordinates. □

Chapter 4

Characters and Their Orthogonality

4.1 Characters

Definition 29 (Character of a representation). The *character* of $V : \mathbf{FDRep} \ \mathbf{k} \ G$ is the function $\chi_V : G \rightarrow k, g \mapsto \mathrm{tr}_V(V \cdot \rho(g))$.

Lemma 30 ($\chi_V(1) = \dim V$). $\chi_V(1) = \dim_k V$.

Proof. $\chi_V(1) = \mathrm{tr}(V \cdot \rho(1)) = \mathrm{tr}(\mathrm{id}_V) = \dim_k V$. This is `FDRep.char_one`. □

Lemma 31 (Characters are class functions). $\chi_V(hgh^{-1}) = \chi_V(g)$ for all $g, h \in G$.

Proof. Cyclicity of trace: $\mathrm{tr}(\rho(h)\rho(g)\rho(h)^{-1}) = \mathrm{tr}(\rho(g))$. Mathlib: `FDRep.char_conj`. □

Lemma 32 (Characters are isomorphism invariants). If $V \cong W$ in $\mathbf{FDRep} \ \mathbf{k} \ G$ then $\chi_V = \chi_W$.

Proof. An isomorphism conjugates one action into the other; trace is conjugation-invariant. Mathlib: `FDRep.char_iso`. □

4.2 The Averaging Formula and Character Orthonormality

Theorem 33 (Average of χ_V counts invariants). For $V : \mathbf{FDRep} \ \mathbf{k} \ G$,

$$\frac{1}{|G|} \sum_{g \in G} \chi_V(g) = \dim_k V^G.$$

Proof. This is `FDRep.average_char_eq_finrank_invariants`. The standard proof: the operator $P := \frac{1}{|G|} \sum_g \rho(g)$ is a G -equivariant idempotent ($P^2 = P$, since left-translation by h permutes the summands and the prefactor is invariant) whose image is exactly V^G . The trace of an idempotent endomorphism of a finite-dimensional vector space is the dimension of its image, giving $\frac{1}{|G|} \sum_g \chi_V(g) = \mathrm{tr}(P) = \dim V^G$. □

Theorem 34 (Hom-space dimension via characters). For $V, W : \mathbf{FDRep} \ \mathbf{k} \ G$,

$$\frac{1}{|G|} \sum_{g \in G} \chi_V(g) \chi_W(g^{-1}) = \dim_k (V \rightarrow W)_G.$$

Proof. Apply Theorem 33 to the representation $\text{Hom}_k(V, W)$ with G -action $g \cdot T := \rho_W(g) T \rho_V(g)^{-1}$. Its character at g equals $\chi_V(g^{-1})\chi_W(g)$, and its G -invariants are exactly $(V \rightarrow W)_G$. Mathlib: `Representation.card_inv_mul_sum_char_mul_char_eq_finrank`. $\square$

Theorem 35 (First Schur orthogonality of characters). *Let $V, W : \mathbf{FDRep} \ \mathbf{k} \ G$ both be simple. Then*

$$\langle \chi_V, \chi_W \rangle_G = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Combine Theorems 34 and 21: the left-hand side equals $\dim_k(V \rightarrow W)_G$ which equals 1 or 0 depending on whether $V \cong W$. Mathlib: `FDRep.char_orthonormal`. $\square$

Chapter 5

Isotypic Decomposition

5.1 The Indexing Set $\hat{G}$

Definition 36 (Set of irreducible classes). Let $\hat{G}$ denote the set of isomorphism classes of simple objects of $\mathbf{FDRep} \mathbf{k} \mathbf{G}$. For $\xi \in \hat{G}$ we choose a representative $V_\xi : \mathbf{FDRep} \mathbf{k} \mathbf{G}$ with $\xi = [V_\xi]$ and write $d_\xi := \dim_k V_\xi$.

Theorem 37 (Finiteness of $\hat{G}$). $\hat{G}$ is finite, with $|\hat{G}| \leq |G|$. (We will see in Theorem 59 that in fact $|\hat{G}|$ equals the number of conjugacy classes of G .)

Proof. Every simple object embeds into the regular representation $k[G]$ (a consequence of Peter–Weyl: see Theorem 45 and the remark there that every irrep appears in $k[G]$ with multiplicity $d_\xi \geq 1$), and $k[G]$ is $|G|$ -dimensional, so $|\hat{G}| \leq |G|$. $\square$

5.2 Isotypic Components

Definition 38 (Isotypic component). For ρ a representation on V and $\xi \in \hat{G}$, the ξ -isotypic component of V is

$$V_{(\xi)} := \sum \{ W \subseteq V : W \text{ is a } G\text{-subspace isomorphic to } V_\xi \},$$

the sum being taken inside V . It is itself a G -subrepresentation.

Lemma 39 (Isotypic components are pairwise disjoint as subspaces). For $\xi \neq \xi'$ in $\hat{G}$, $V_{(\xi)} \cap V_{(\xi')} = 0$ inside V .

Proof. Write $W := V_{(\xi)} \cap V_{(\xi')}$, a G -subrepresentation of V . By Maschke (Theorem 19) W is semisimple, i.e. a direct sum of simples. Each simple summand sits in both $V_{(\xi)}$ and $V_{(\xi')}$, so by the definition of isotypic component is isomorphic to both V_ξ and $V_{\xi'}$, contradicting $\xi \neq \xi'$. Hence there are no simple summands and $W = 0$. $\square$

Lemma 40 (Multiplicity space). For $V : \mathbf{FDRep} \mathbf{k} \mathbf{G}$ and $\xi \in \hat{G}$, define $M_\xi := (V_\xi \rightarrow V)_G$ (the k -vector space of G -equivariant maps $V_\xi \rightarrow V$). Then $\dim_k M_\xi$ is the multiplicity m_ξ of V_ξ in V , and there is a canonical G -equivariant isomorphism

$$\mathrm{ev}_\xi : V_\xi \otimes_k M_\xi \xrightarrow{\sim} V_{(\xi)}, \quad v \otimes \phi \mapsto \phi(v).$$

In particular $V_{(\xi)} \cong V_\xi^{\oplus m_\xi}$ and $\dim_k V_{(\xi)} = m_\xi \cdot d_\xi$.

Proof. Equivariance of ev_ξ is direct. Surjectivity: by Theorem 19, $V_{(\xi)}$ is a finite direct sum $\bigoplus_\alpha W_\alpha$ of simples isomorphic to V_ξ ; pick isomorphisms $\phi_\alpha : V_\xi \rightarrow W_\alpha \subseteq V$, then any $w \in V_{(\xi)}$ writes as $\sum_\alpha \phi_\alpha(v_\alpha) = \text{ev}_\xi(\sum_\alpha v_\alpha \otimes \phi_\alpha)$. Injectivity: an element of $V_\xi \otimes M_\xi$ in the kernel decomposes as $\sum v_\alpha \otimes \phi_\alpha$ for ϕ_α a basis of M_ξ ; the requirement $\sum \phi_\alpha(v_\alpha) = 0$ together with Theorem 21 (which says the ϕ_α have linearly independent images among simples) forces every $v_\alpha = 0$. The dimension count follows. $\square$

Theorem 41 (Isotypic decomposition). *For every $V : \text{FDRep } \mathbf{k} \mathbf{G}$, the inclusions $V_{(\xi)} \hookrightarrow V$ assemble into a G -equivariant isomorphism*

$$\bigoplus_{\xi \in \hat{G}} V_{(\xi)} \xrightarrow{\sim} V,$$

and consequently

$$V \cong \bigoplus_{\xi \in \hat{G}} V_\xi^{\oplus m_\xi}, \quad \dim_k V = \sum_{\xi \in \hat{G}} m_\xi d_\xi.$$

Proof. By Theorem 19 write $V = \bigoplus_\alpha W_\alpha$ with each W_α simple. Group the summands by their isomorphism class $\xi \in \hat{G}$; the sum of all W_α in class ξ is exactly $V_{(\xi)}$ (it is contained in $V_{(\xi)}$ and contains every simple summand of V isomorphic to V_ξ). The $V_{(\xi)}$ jointly span V by construction, and pairwise intersect trivially by Lemma 39; hence they form an internal direct sum decomposition of V . Combine with the description from Lemma 40. $\square$

Chapter 6

The Peter–Weyl Theorem for Finite Groups

6.1 The Regular Representation

Definition 42 (Left regular representation). The *left regular representation* is $\lambda : G \rightarrow (k[G] \rightarrow_{\ell[k]} k[G])$ given by $\lambda(g)(\delta_h) = \delta_{gh}$ on the basis $\{\delta_h : h \in G\}$. Mathlib: `Representation.leftRegular k G` (and its bundled form `Rep.ofMulAction k G G : Rep k G`).

Lemma 43 (Character of the regular representation). $\chi_{k[G]}(g) = |G|$ if $g = 1$, and 0 otherwise.

Proof. The matrix of $\lambda(g)$ in the basis $\{\delta_h\}_{h \in G}$ is the permutation matrix of left translation by g . Its trace is the number of fixed points of $h \mapsto gh$, which is $|G|$ if $g = 1$ and 0 otherwise (a free G -action on G has no fixed points unless $g = 1$). $\square$

6.2 Multiplicities in $k[G]$

Lemma 44 (Multiplicity of V_ξ in $k[G]$). Let $\xi \in \hat{G}$ and m_ξ denote the multiplicity of V_ξ in $k[G]$ (Lemma 40). Then $m_\xi = d_\xi$.

Proof. By Lemma 40, $m_\xi = \dim_k(V_\xi \rightarrow k[G])_G$. By Theorem 34,

$$\dim_k(V_\xi \rightarrow k[G])_G = \frac{1}{|G|} \sum_g \chi_{V_\xi}(g) \chi_{k[G]}(g^{-1}).$$

Using Lemma 43 only the $g = 1$ term survives, giving $\frac{1}{|G|} \cdot \chi_{V_\xi}(1) \cdot |G| = \chi_{V_\xi}(1) = d_\xi$ by Lemma 30. $\square$

6.3 The Peter–Weyl Decomposition

Theorem 45 (Peter–Weyl: decomposition of the regular representation). As k -linear G -representations,

$$k[G] \cong \bigoplus_{\xi \in \hat{G}} V_\xi^{\oplus d_\xi}.$$

In particular every irreducible representation appears in $k[G]$ with multiplicity equal to its dimension.

Proof. Apply Theorem 41 to $V := k[G]$, using Lemma 44 to identify each multiplicity m_ξ with d_ξ . $\square$

Corollary 46 (Sum of squares of dimensions). $|G| = \sum_{\xi \in \hat{G}} d_\xi^2$.

Proof. Take $\dim_k$ of both sides of Theorem 45: the left side is $|G|$, the right side is $\sum_\xi d_\xi \cdot d_\xi = \sum_\xi d_\xi^2$. $\square$

6.4 Two-Sided Wedderburn Form (optional)

Theorem 47 (Wedderburn form of the group algebra). *As k -algebras (equivalently, as two-sided $k[G]$ -modules) there is an isomorphism*

$$k[G] \cong \prod_{\xi \in \hat{G}} \text{End}_k(V_\xi).$$

Proof. The k -algebra map $\Phi : k[G] \rightarrow \prod_\xi \text{End}_k(V_\xi)$ sending $\sum_g a_g g \mapsto (\sum_g a_g V_\xi \cdot \rho(g))_\xi$ is well-defined and an algebra homomorphism by definition of $V_\xi \cdot \rho$ as a monoid hom. Surjectivity: under the equivalence `Rep.equivalenceModuleMonoidAlgebra`, Φ corresponds to the natural map $k[G] \rightarrow \text{End}_{k[G]}(\bigoplus_\xi V_\xi^{\oplus d_\xi}) \cong \prod_\xi \text{End}_k(V_\xi)^{\text{op}}$ (the second isomorphism uses Corollary 22 and the Schur identification $\text{End}_{k[G]}(V_\xi^{\oplus d_\xi}) \cong M_{d_\xi}(k)$). Injectivity: source and target have equal dimension over k , namely $|G| = \sum_\xi d_\xi^2$ (Corollary 46); since Φ is surjective and source and target are finite-dimensional with matching dimension, Φ is bijective. $\square$

Chapter 7

Multiplicity Formula and Character Theory Consequences

7.1 Multiplicity from Characters

Theorem 48 (Multiplicity formula). *For any $V : \mathbf{FDRep} \ k \ G$ and $\xi \in \hat{G}$, the multiplicity m_ξ of V_ξ in V satisfies*

$$m_\xi = \langle \chi_V, \chi_{V_\xi} \rangle_G = \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \chi_{V_\xi}(g^{-1}).$$

Proof. By Lemma 40, $m_\xi = \dim_k(V_\xi \rightarrow V)_G$. By Theorem 34, $\dim_k(V_\xi \rightarrow V)_G = |G|^{-1} \sum_g \chi_{V_\xi}(g) \chi_V(g^{-1})$. Replacing $g \mapsto g^{-1}$ (a bijection of G) shows this equals $|G|^{-1} \sum_g \chi_V(g) \chi_{V_\xi}(g^{-1}) = \langle \chi_V, \chi_{V_\xi} \rangle_G$. $\square$

Corollary 49 (Characters separate isomorphism classes). *For $V, W : \mathbf{FDRep} \ k \ G$, $V \cong W$ iff $\chi_V = \chi_W$.*

Proof. $(\Rightarrow)$ is Lemma 32. $(\Leftarrow)$: by Theorem 48, $\chi_V = \chi_W$ implies the multiplicities m_ξ for V and W coincide; by Theorem 41 the multiplicities determine the isomorphism class, so $V \cong W$. $\square$

Corollary 50 (Irreducibility test). *$V : \mathbf{FDRep} \ k \ G$ is simple iff $\langle \chi_V, \chi_V \rangle_G = 1$.*

Proof. Write $V \cong \bigoplus_\xi V_\xi^{\oplus m_\xi}$. Then $\chi_V = \sum_\xi m_\xi \chi_{V_\xi}$ and by orthonormality (Theorem 35), $\langle \chi_V, \chi_V \rangle_G = \sum_\xi m_\xi^2$, an integer. This equals 1 iff exactly one m_ξ is 1 and all others are 0, iff $V \cong V_\xi$ is simple. $\square$

7.2 Class Functions and Their Span

Definition 51 (Space of class functions). A *class function* on G is a function $f : G \rightarrow k$ with $f(hgh^{-1}) = f(g)$ for all g, h . Class functions form a k -subspace $\text{Cl}(G) \subseteq (G \rightarrow k)$ whose dimension equals the number of conjugacy classes of G (with delta functions on conjugacy classes as a basis).

Lemma 52 ($\dim_k \text{Cl}(G) = \text{number of conjugacy classes}$). $\dim_k \text{Cl}(G) = |\text{ConjClasses } G|$.

Proof. A class function is determined by, and freely chosen on, each conjugacy class. The indicator functions of conjugacy classes form a basis. $\square$

Lemma 53 (Characters lie in $\text{Cl}(G)$). *For every $V : \text{FDRep } \mathbf{k} \ G$, $\chi_V \in \text{Cl}(G)$.*

Proof. Lemma 31. $\square$

7.3 Irreducible Characters Span $\text{Cl}(G)$

The next theorem is the second main result of finite-group character theory after the Peter–Weyl decomposition. We isolate the spanning argument as a separate lemma to enable independent formalisation.

Lemma 54 (The projection T_f associated to $f \in \text{Cl}(G)$). *For $f \in \text{Cl}(G)$ and ρ a representation of G on V , define*

$$T_f^\rho := \frac{1}{|G|} \sum_{g \in G} f(g) \rho(g) \in (V \rightarrow_{\ell[k]} V).$$

Then T_f^ρ is G -equivariant: $\rho(h) T_f^\rho = T_f^\rho \rho(h)$ for every h .

Proof. Compute $\rho(h) T_f^\rho \rho(h)^{-1} = |G|^{-1} \sum_g f(g) \rho(hgh^{-1}) = |G|^{-1} \sum_{g'} f(h^{-1}g'h) \rho(g') \ (g' := hgh^{-1}) = |G|^{-1} \sum_{g'} f(g') \rho(g') \text{ (using } f \text{ is a class function)} = T_f^\rho$. $\square$

Lemma 55 (Schur scalar of T_f on a simple). *For $\xi \in \hat{G}$, $T_f^{V_\xi} = \mu_\xi(f) \text{id}_{V_\xi}$ where*

$$\mu_\xi(f) = \frac{1}{d_\xi \cdot |G|} \sum_{g \in G} f(g) \chi_{V_\xi}(g).$$

In particular $T_f^{V_\xi} = 0$ iff $\sum_g f(g) \chi_{V_\xi}(g) = 0$.

Proof. By Lemma 54 $T_f^{V_\xi}$ is in $\text{End}_G(V_\xi)$, which by Corollary 22 is one-dimensional spanned by id_{V_ξ} . So $T_f^{V_\xi} = \mu_\xi(f) \text{id}_{V_\xi}$. Take k -trace of both sides: $\text{tr}(T_f^{V_\xi}) = |G|^{-1} \sum_g f(g) \chi_{V_\xi}(g)$ on the left and $\mu_\xi(f) \cdot d_\xi$ on the right; solve for $\mu_\xi(f)$. $\square$

Lemma 56 (f orthogonal to all χ_{V_ξ} implies $T_f^V = 0$ for every V). *Let $f \in \text{Cl}(G)$. If $\sum_g f(g) \chi_{V_\xi}(g) = 0$ for every $\xi \in \hat{G}$, then for every $V : \text{FDRep } \mathbf{k} \ G$, $T_f^V = 0$.*

Proof. By Theorem 41, V is a direct sum of simples V_ξ , and T_f^V acts diagonally on this decomposition by Lemma 54. By Lemma 55 the hypothesis implies $T_f^{V_\xi} = 0$ on each summand, hence $T_f^V = 0$. $\square$

Lemma 57 (Applying T_f on the regular representation to δ_1). *Let $\rho = \lambda$ be the regular representation on $V = k[G]$, and let $\delta_1 \in k[G]$ denote the basis vector at $1 \in G$. Then $T_f^\lambda(\delta_1) = \frac{1}{|G|} \sum_{g \in G} f(g) \delta_g = \frac{1}{|G|} f$, where on the right we identify $f \in (G \rightarrow k)$ with the element $\sum_g f(g) \delta_g \in k[G]$.*

Proof. Compute $T_f^\lambda(\delta_1) = |G|^{-1} \sum_g f(g) \lambda(g) \delta_1 = |G|^{-1} \sum_g f(g) \delta_g$. Identifying δ_g with the basis function $\mathbf{1}_{\{g\}} \in (G \rightarrow k)$ and recombining gives $|G|^{-1} f$. $\square$

Theorem 58 (Irreducible characters span $\text{Cl}(G)$). *The irreducible characters $\{\chi_{V_\xi}\}_{\xi \in \hat{G}}$ span $\text{Cl}(G)$.*

Proof. Suppose $f \in \text{Cl}(G)$ satisfies $\sum_g f(g)\chi_{V_\xi}(g) = 0$ for every $\xi \in \hat{G}$ (this is equivalent to $\langle f, \chi_{V_\xi^*} \rangle_G = 0$ for every ξ , where V_ξ^* is the dual representation; since $\xi \mapsto \xi^*$ permutes $\hat{G}$, this is the same as orthogonality to all irreducible characters in $\langle \cdot, \cdot \rangle_G$). By Lemma 56 applied to the regular representation, $T_f^\lambda = 0$. Apply this to δ_1 : Lemma 57 gives $0 = T_f^\lambda(\delta_1) = |G|^{-1} f$ in $k[G]$, hence $f = 0$ in k^G . So the orthogonal complement of the irreducible characters in $\text{Cl}(G)$ is zero, i.e. they span. $\square$

Theorem 59 (Irreducible characters are an orthonormal basis of $\text{Cl}(G)$). *$\{\chi_{V_\xi}\}_{\xi \in \hat{G}}$ is an orthonormal basis of $\text{Cl}(G)$ for the inner product $\langle \cdot, \cdot \rangle_G$. In particular,*

$$|\hat{G}| = \dim_k \text{Cl}(G) = |\text{ConjClasses } G|.$$

Proof. Orthonormality is Theorem 35, hence the χ_{V_ξ} are linearly independent. Spanning is Theorem 58. Hence they form a basis. The cardinality identity $|\hat{G}| = |\text{ConjClasses } G|$ follows by comparing $\dim_k \text{Cl}(G)$ via this basis with Lemma 52. $\square$

Chapter 8

Summary of Mathlib Status

Already in Mathlib ([ML]). The following declarations underpin the entire blueprint: `Representation`, `Representation.asModule`, `Rep`, `FDRep`, `CategoryTheory.Simple`, `Representation.invariants`, `LinearMap.conjugate`, `LinearMap.conjugate_i`, `LinearMap.sumOfConjugatesEquivariant`, `LinearMap.equivariantProjection`, `LinearMap.equivariantProjection_condition`, `MonoidAlgebra.exists_le`, `MonoidAlgebra.Submodule.exists_isCompl`, `MonoidAlgebra.Submodule.instIsSemisimpleModule`, `FDRep.finrank_hom_simple_simple (Schur)`, `FDRep.character`, `FDRep.char_one`, `FDRep.char_conj`, `FDRep.char_iso`, `FDRep.char_orthonormal`, `FDRep.average_char_eq_finrank_invariants`, `Representation.card_inv_mul_sum_char_mul_char_eq_finrank`, `Representation.leftRegular`, `Rep.ofMulAction`.

Close analogues ([~ML]). The space `ClassFunction` and the dimension count $\dim \text{Cl}(G) = |\text{ConjClasses } G|$ are not packaged but trivial. The character of the regular representation $(\chi_{k[G]} = |G| \cdot \mathbf{1}_{\{1\}})$ is folklore; trivial from the permutation-matrix description of `Rep.ofMulAction`.

Newly required ([NEW]). The contributions to Mathlib that this blueprint amounts to are: the inner product $\langle \cdot, \cdot \rangle_G$ (Definition 10), the symmetrisation (Definition 23) and matrix-coefficient orthogonality (Theorems 27, 28), the indexing set $\hat{G}$ and its finiteness, the isotypic component and multiplicity space (Definition 38, Lemma 40), the isotypic decomposition (Theorem 41), the Peter–Weyl decomposition (Theorem 45) and its sum-of-squares corollary, the optional Wedderburn form (Theorem 47), the multiplicity formula (Theorem 48) and its corollaries, and the spanning of $\text{Cl}(G)$ by irreducible characters (Theorem 58, Theorem 59).